Hemaang Sood

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Github: hemaangsood | Portfolio: hemaangsood.github.io

Skills

- **AI, ML & LLM Orchestration:** PyTorch, OpenAI API, LangChain, DSPy, RAG, OpenCV, vLLM, Ray (Ray Serve).
- **Programming Languages:** Python, JavaScript, Go, Kotlin.
- **Backend & Web Development:** FastAPI, Django, Spring Boot, NodeJS, React.js, Tailwind CSS.
- **Data, Retrieval & Databases:** SQL, PostgreSQL, MongoDB.
- **Cloud, DevOps & Tools:** AWS (EC2), Docker, Nginx, Git, GitHub.

Experience

University of Waterloo Waterloo, ON
ECE 699 Research — Multi-Agent LLM Optimization with DSPy and GEPA Sept 2026 – Present

- Research project optimizing **multi-agent LLM coordination** by evolving agent prompts with **GEPA** on **DSPy**, benchmarked on Hanabi, SciBench, and maze navigation.
- Built a **16-GPU AMD Instinct MI50** (ROCm) cluster serving **vLLM** tensor-parallel models behind a **Ray Serve** router; delivered a 4–12x speed-up and 6x more evaluations on the same hardware without starvation.

Synechron Bangalore, India
Jr Associate May 2025 – Nov 2025

- Built a full-stack **Amusement Park Management System** using **MERN** with **Vite.js** and **Tailwind CSS**, focusing on modular UI design, API integration, and database connectivity.
- Built a backend server for an **Insurance Management System** using **Spring Boot**, focusing on scalability, security, and modularity.

Bharat Electronics Limited (BEL) Bangalore, India
Intern April 2024 – Nov 2024

- Developed an Authentication and Authorization module for a secure **DRDO** intranet website.
- Built an **LLM** interface using **RAG** to process and extract insights from sensitive documents; integrated a **Vosk** voice-input pipeline to transcribe speech into queries.

Education

University of Waterloo Jan 2026 – April 2027
Electrical & Computer Engineering, MEng GPA - 3.9 / 4.0

BMS Institute of Technology and Management (BMSIT&M) 2021 – 2025
Bachelor of Engineering in Computer Science and Engineering CGPA - 8.9 / 10.0

Conference Publications & Presentations

- Hemaang Sood et al., "Improving Face Recognition for Smart Home Security: A Study of ResNet, Siamese Networks, and Attention Mechanisms" Presented at IEEE COMPSIF 2025.

Personal Projects

Virtual POCT — AI-Powered Quality Control Assistant: *Individual*

- Built a full-stack AI-powered quality control platform for Point-of-Care Testing with a conversational assistant and auditable QC workflows.
- Designed a streaming WebSocket architecture (**FastAPI**, **LangChain**) for real-time LLM responses; delivered a Dockerized microservice with JWT auth and RBAC.
- Implemented a deterministic quality control evaluation engine that combines operator input with database context to classify QC scenarios and automate compliance actions.
- **Technologies Used:** FastAPI, React, TypeScript, PostgreSQL, SQLAlchemy, Docker, Nginx, WebSockets, LangChain, Google Gemini API, JWT.

Facial and Voice Recognition-Based Door Lock System: *Group*

- Achieved 98.7% accuracy by combining facial and speech recognition (custom YOLO model + Siamese Network on the YaleFace dataset).
- Published paper in **IEEE COMPSIF 2025** for this project.
- **Technologies and Tools:** OpenCV, PyTorch, Google Speech API, Python, Raspberry Pi, Docker, React, PostgreSQL, Tailwind, AWS EC2, Django, NGINX.

Routed: Travel Partner Recommendation System *Group*

- Built a microservices platform matching travelers via trip overlap, preferences, and geospatial proximity (FastAPI, JWT, PostgreSQL/PostGIS).
- Built an event-driven pipeline with **Kafka** and a **Go** worker (MongoDB) plus a **Kotlin** microservice for semantic embeddings.
- **Technologies Used:** FastAPI, React, TypeScript, PostgreSQL, PostGIS, Kafka, MongoDB, Docker, Go, Kotlin, SQLAlchemy.

Resume Builder: *Individual*

- Built an AI-powered tool that parses LaTeX resumes and tailors content to job descriptions via a multi-step agent.
- **Technologies Used:** Python, OpenAI API, LaTeX, SQLite, Jinja2, pdflatex.